



NETWORK
FORENSICS



WIRESHARK
ANALYSIS



MALWARE
DETECTION



THREAT
INTELLIGENCE

ADVANCED MALWARE TRAFFIC ANALYSIS AND INCIDENT INVESTIGATION REPORT



Wireshark-Based
Network Forensics Investigation



PACKET
ANALYSIS



MALWARE
TRAFFIC



THREAT
DETECTION



INCIDENT
INVESTIGATION



FORENSIC
EVIDENCE

WIRESHARK

No.	Time	Source	Destination	Protocol	Length	Info
2456	12.451528	192.168.1.25	185.199.108.153	TCP	54	51234 → 443 [SYN]
2457	12.451661	185.199.108.153	192.168.1.25	TCP	54	443 → 51234 [SYN, ACK]
2458	12.451789	192.168.1.25	185.199.108.153	TCP	54	51234 → 443 [ACK]
2461	12.452111	192.168.1.25	45.135.23.67	HTTP	512	GET /malware.exe
2462	12.452776	45.135.23.67	192.168.1.25	HTTP	1460	200 OK (application/x-ms.1
2463	12.453443	192.168.1.25	45.135.23.67	TCP	1460	[TCP segment of a reassembl
2464	12.454100	192.168.1.25	8.8.8.8	DNS	86	Standard query 0x6a2f A .1
2465	12.454881	8.8.8.8	192.168.1.25	DNS	102	Standard query response .1

TCP Stream Graph



splunk>enterprise



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Wireshark Packet Analysis



Malware Traffic Investigation



Network Forensics & Threat Hunting



Incident Response & Correlation



Digital Evidence Collection



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EXECUTIVE SUMMARY

This investigation focuses on the analysis of suspicious network traffic captured within a packet capture (PCAP) file using Wireshark as part of a practical malware traffic analysis exercise.

The primary objective of this investigation was to identify suspicious network behavior, malicious communications, unusual DNS activity, abnormal HTTP requests, and indicators of compromise associated with potential malware infection.

During the investigation, detailed traffic inspection revealed that internal host 10.1.17.215 communicated with suspicious external infrastructure, including IP address 5.252.153.241, which demonstrated malicious behavior through repeated outbound HTTP communication.

Multiple suspicious HTTP GET requests were identified, including requests for PowerShell script payloads and executable file retrieval activity, strongly suggesting malware delivery and execution behavior.

Threat intelligence validation using VirusTotal confirmed that the identified external IP address was flagged as malicious by multiple security vendors, increasing confidence in the malicious classification of the observed traffic.

Based on the collected evidence, this investigation demonstrates characteristics consistent with malware infection, suspicious payload delivery, and external command-style communication, reflecting real-world SOC and incident response analysis workflows.

OBJECTIVE

The objective of this project was to perform practical malware traffic analysis on a suspicious PCAP file using Wireshark to identify malicious network activity, suspicious communications, malware delivery behavior, and indicators of compromise through packet-level forensic investigation.

- Identify the infected internal host system
- Detect suspicious DNS communication activity
- Analyze malicious HTTP traffic and payload delivery attempts
- Identify suspicious external IP communication
- Validate malicious indicators using threat intelligence sources
- Extract indicators of compromise (IOCs)
- Understand malware communication behavior
- Build a professional SOC-style incident investigation report

TOOLS USED

Tool	Purpose
Wireshark	Network packet analysis
VirusTotal	Threat intelligence validation
Malware-Traffic-Analysis.net	PCAP dataset source
Windows Analysis Environment	Investigation platform

INVESTIGATION METHODOLOGY

The investigation began by loading the suspicious PCAP file into Wireshark for detailed packet-level analysis. Initial review confirmed that the capture contained approximately 39,427 packets, providing sufficient network evidence for forensic examination. To identify suspicious internal activity, display filters were applied to isolate abnormal communications. Through this process, internal host 10.1.17.215 was identified as the primary system generating suspicious traffic and became the focus of the investigation. This phase established the foundation for further malware traffic analysis and incident investigation.

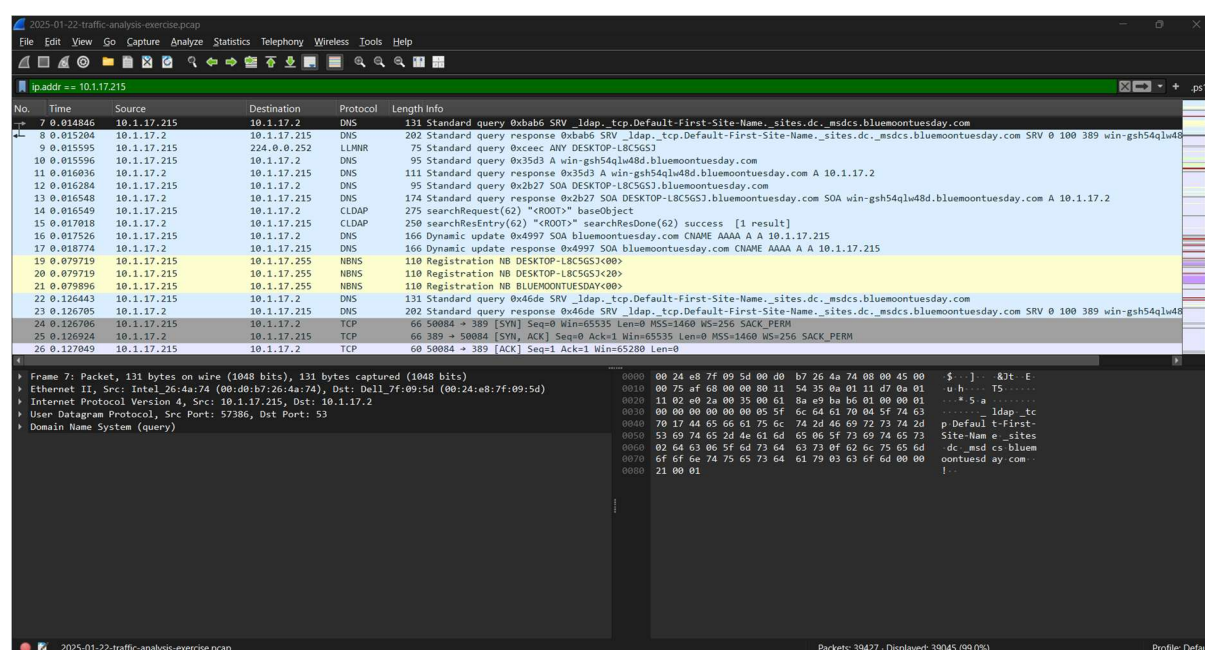


Figure 1: Network traffic generated by suspected infected internal host (10.1.17.215)

DNS TRAFFIC ANALYSIS

Following internal host identification, DNS traffic analysis was performed to examine domain resolution activity generated by the suspected infected system. DNS packet inspection revealed repeated outbound domain queries, including communication with suspicious external infrastructure. During analysis, the domain ping3.dyngate.com was identified, which appeared abnormal within the context of this investigation and indicated possible malware-related beaconing or external callback communication behavior. DNS analysis played an important role in identifying suspicious network activity before deeper HTTP traffic inspection.

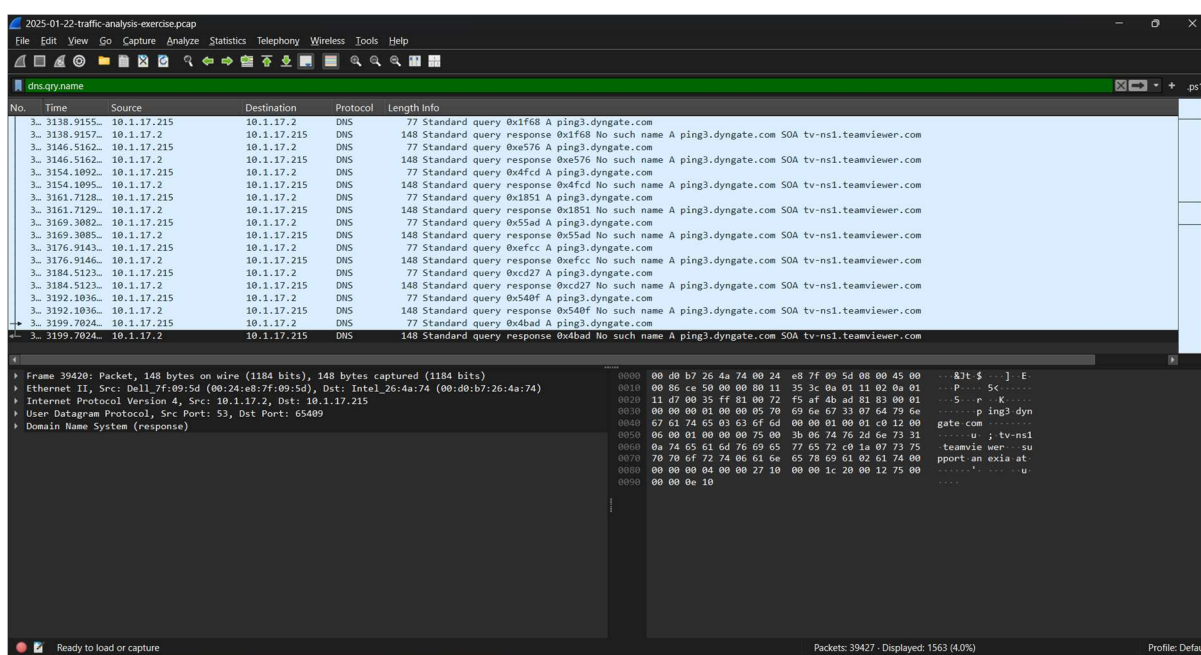
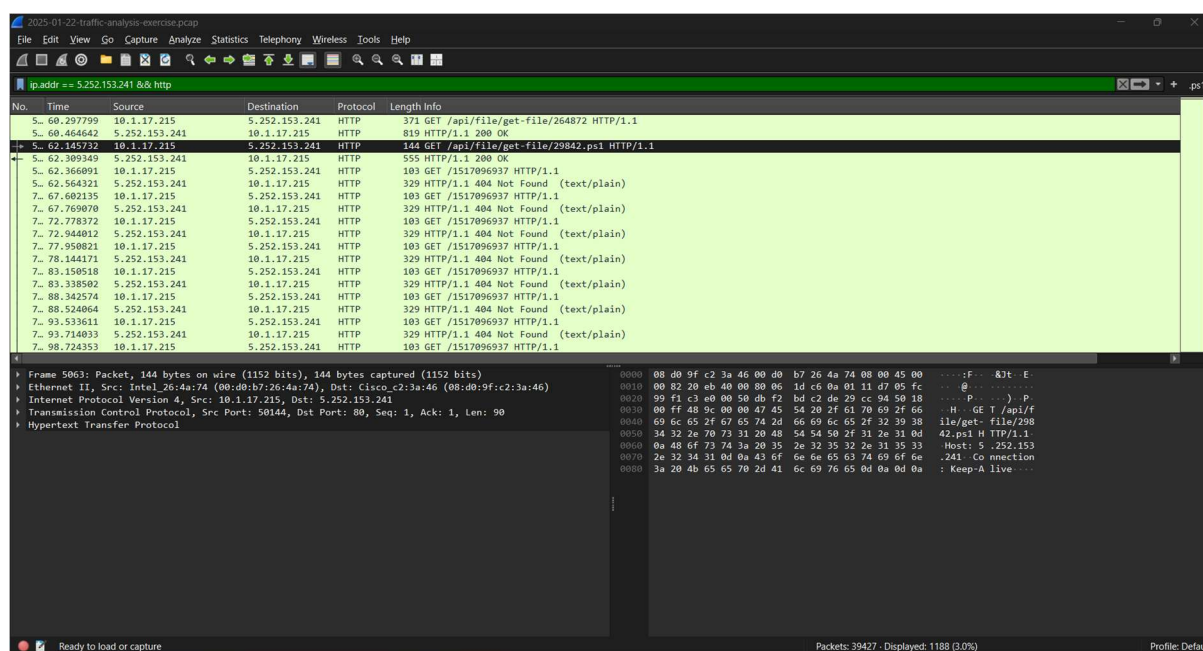


Figure 2: Suspicious DNS queries revealing abnormal external domain resolution activity

HTTP MALICIOUS TRAFFIC ANALYSIS

Following DNS investigation, HTTP traffic analysis was performed to inspect outbound web communication generated by the suspected infected system. Examination of HTTP requests revealed suspicious file retrieval behavior between internal host 10.1.17.215 and external IP address 5.252.153.241. Multiple abnormal GET requests were identified, including direct requests for downloadable files and PowerShell script payload retrieval activity. The presence of a .ps1 file request strongly indicates malware staging, payload delivery, or malicious script execution behavior commonly observed during malware infections and command-and-control operations.



No.	Time	Source	Destination	Protocol	Length	Info
5...	60.297799	10.1.17.215	5.252.153.241	HTTP	371	GET /api/file/get-file/264872 HTTP/1.1
5...	60.464642	5.252.153.241	10.1.17.215	HTTP	819	HTTP/1.1 200 OK
5...	62.145732	10.1.17.215	5.252.153.241	HTTP	144	GET /api/file/get-file/29842.ps1 HTTP/1.1
5...	62.309349	5.252.153.241	10.1.17.215	HTTP	555	HTTP/1.1 200 OK
5...	62.366091	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1
5...	62.564321	5.252.153.241	10.1.17.215	HTTP	329	HTTP/1.1 404 Not Found (text/plain)
7...	67.602135	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1
7...	67.769070	5.252.153.241	10.1.17.215	HTTP	329	HTTP/1.1 404 Not Found (text/plain)
7...	72.778372	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1
7...	72.944012	5.252.153.241	10.1.17.215	HTTP	329	HTTP/1.1 404 Not Found (text/plain)
7...	77.950821	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1
7...	78.144171	5.252.153.241	10.1.17.215	HTTP	329	HTTP/1.1 404 Not Found (text/plain)
7...	83.150518	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1
7...	83.338502	5.252.153.241	10.1.17.215	HTTP	329	HTTP/1.1 404 Not Found (text/plain)
7...	88.342574	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1
7...	88.524064	5.252.153.241	10.1.17.215	HTTP	329	HTTP/1.1 404 Not Found (text/plain)
7...	93.533611	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1
7...	93.714033	5.252.153.241	10.1.17.215	HTTP	329	HTTP/1.1 404 Not Found (text/plain)
7...	98.724353	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1

Frame 5063: Packet, 144 bytes on wire (1152 bits), 144 bytes captured (1152 bits)	Details	Bytes
Ethernet II, Src: Intel_26:4a:74 (00:d0:b7:26:4a:74), Dst: Cisco_c2:3a:46 (08:d0:9f:c2:3a:46)		0000 08 d0 9f c2 3a 46 00 d0 b7 26 4a 74 08 00 45 00
Internet Protocol Version 4, Src: 10.1.17.215, Dst: 5.252.153.241		0010 00 82 20 eb 40 00 00 06 1d c6 0a 01 11 d7 05 fc
Transmission Control Protocol, Src Port: 50144, Dst Port: 80, Seq: 1, Ack: 90		0020 90 f1 c3 e0 00 50 db f2 bd c2 de 29 cc 94 50 18
Hypertext Transfer Protocol		0030 00 ff 48 9c 00 00 47 45 54 20 2f 61 70 69 2f 66
		0040 69 6c 65 2f 67 65 74 2d 66 69 6c 65 2f 32 39 38
		0050 34 32 2e 70 73 31 20 48 54 54 50 2f 31 2e 31 0d
		0060 0a 48 6f 73 74 3a 20 35 2e 32 35 32 2e 31 35 33
		0070 2e 32 34 31 0d 0a 43 6f 6e 6e 63 74 69 6f 6e
		0080 3a 20 4b 65 65 70 2d 41 6c 69 76 65 0d 0a 0d 0a

Figure 3: Suspicious HTTP communication showing PowerShell payload retrieval from malicious external infrastructure

EXTERNAL MALICIOUS IP COMMUNICATION ANALYSIS

To further validate suspicious activity, communication with the external IP address 5.252.153.241 was isolated and analyzed in detail. Packet inspection revealed repeated bidirectional communication between the internal host and the suspicious external infrastructure. This communication included multiple HTTP transactions and abnormal data exchange behavior. The repeated interaction pattern strongly suggested malware callback communication, payload transfer, or command-and-control style network activity. This phase helped establish a direct relationship between the infected internal machine and suspicious external infrastructure.

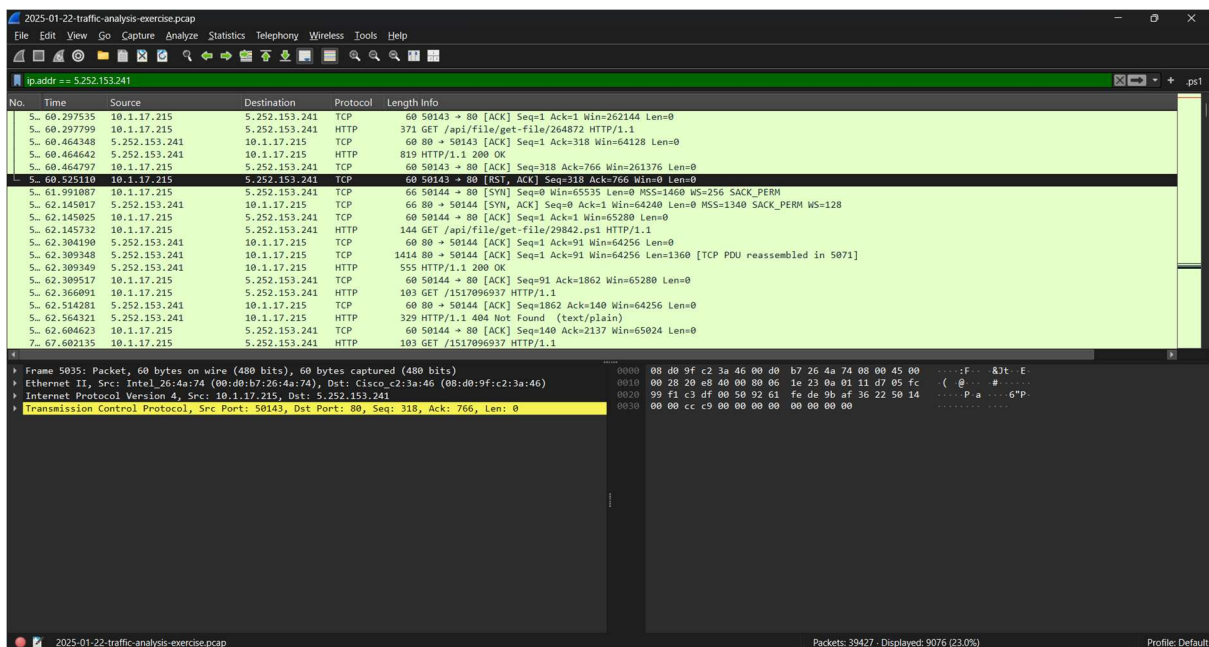


Figure 4: Direct communication between infected internal host and suspicious malicious external infrastructure

TCP CONVERSATION ANALYSIS

To measure the extent of malicious communication, TCP conversation analysis was performed using Wireshark statistical conversation tracking. The analysis revealed significant bidirectional communication between internal host 10.1.17.215 and suspicious external IP address 5.252.153.241. Traffic statistics showed thousands of packets exchanged along with approximately 5 MB of transferred data, indicating sustained communication rather than isolated activity. Such behavior strongly suggests malware payload transfer, beaconing activity, or command-and-control communication between the compromised endpoint and malicious external infrastructure.

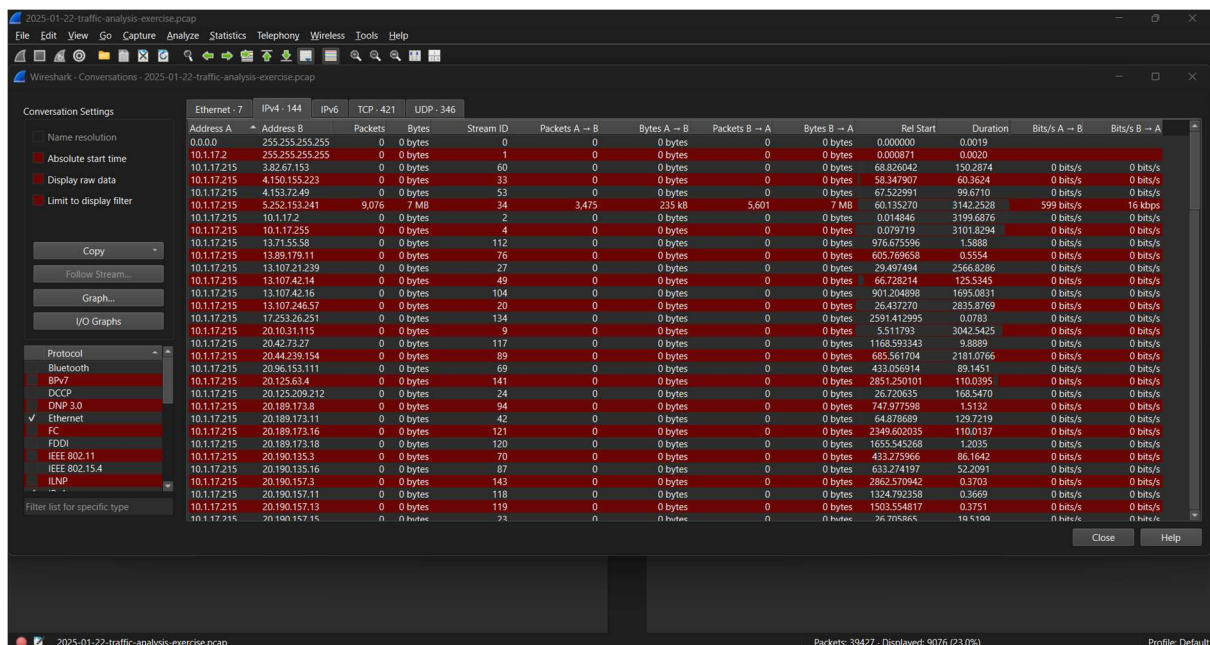


Figure 5: TCP conversation statistics showing high-volume communication with malicious infrastructure

THREAT INTELLIGENCE VALIDATION

To validate the suspicious external infrastructure identified during packet analysis, threat intelligence verification was performed using VirusTotal. The external IP address 5.252.153.241 was analyzed against multiple security intelligence engines. Results confirmed that multiple security vendors classified the IP address as malicious, including phishing and malware-related detections. This external validation significantly increased confidence in the investigation findings and confirmed that the internal host was communicating with known malicious infrastructure. Threat intelligence correlation is a critical SOC investigation practice for confirming indicators of compromise.

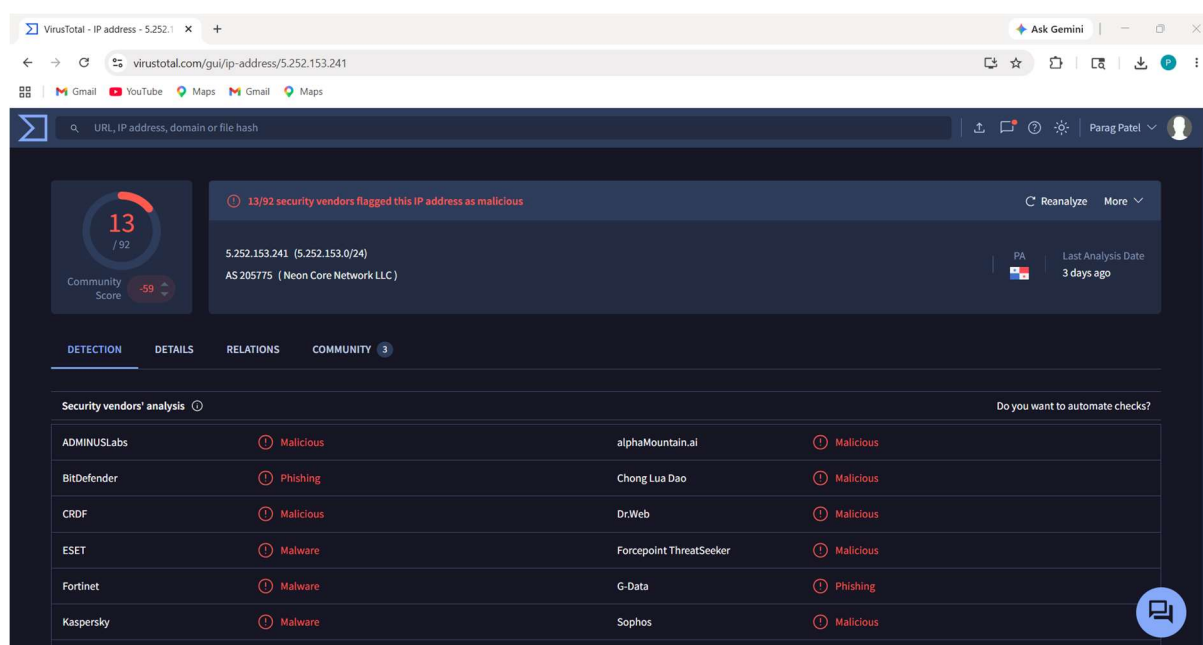


Figure 6: VirusTotal threat intelligence validation confirming malicious classification of suspicious external IP

INDICATORS OF COMPROMISE (IOCs)

Indicator Type	Value	Description
Infected Internal Host	10.1.17.215	Primary compromised internal endpoint identified during analysis
Malicious External IP	5.252.153.241	Suspicious external infrastructure communicating with infected host
Suspicious Domain	ping3.dyngate.com	Abnormal DNS query observed during malware traffic analysis
Malicious Payload	29842.ps1	PowerShell script payload requested via HTTP
Suspicious Protocols	DNS, HTTP, TCP	Protocols involved in malicious communication

Malicious HTTP Payload Retrieval

During HTTP traffic inspection, the infected host **10.1.17.215** initiated outbound communication with suspicious external IP **5.252.153.241** over port 80. Analysis revealed HTTP GET requests attempting to retrieve potentially malicious content, including a suspicious PowerShell payload named **29842.ps1**. PowerShell-based payload delivery is a commonly used malware execution technique due to its native Windows availability and scripting capabilities. This strongly indicates malware payload download behavior and active compromise.

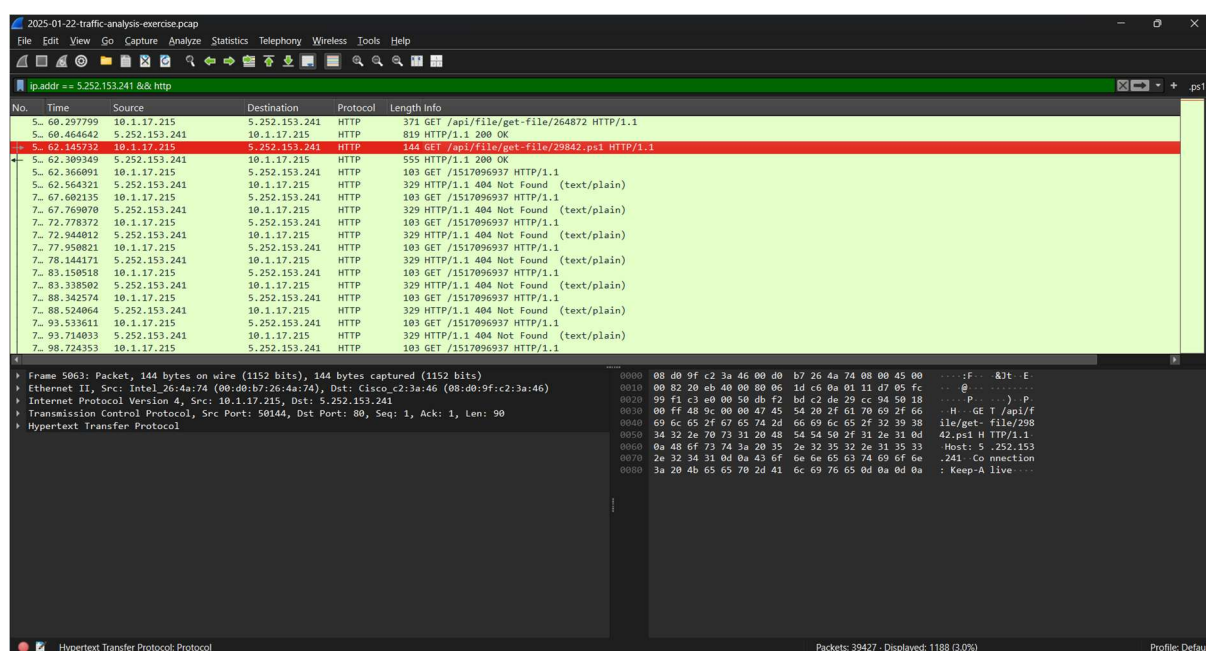


Figure 7: HTTP GET request retrieving suspicious PowerShell payload

ATTACK CHAIN ANALYSIS

Based on the observed network evidence, the attack appears to follow a structured malware infection lifecycle commonly seen in real-world cyber incidents.

Attack Sequence Identified:

- Initial compromise of internal host 10.1.17.215
- Suspicious DNS resolution activity toward external infrastructure
- Outbound HTTP communication establishment with suspicious IP address 5.252.153.241
- Retrieval of suspicious PowerShell payload (29842.ps1)
- Continued repeated communication indicating persistent malware activity
- Large-volume TCP communication suggesting payload transfer or command-and-control behavior
- External infrastructure validated as malicious through threat intelligence analysis

MITRE ATT&CK TECHNIQUE MAPPING

Observed malware behavior during the investigation aligns with multiple MITRE ATT&CK adversarial techniques commonly observed in enterprise cyber incidents.

Mapped Techniques:

- **T1071 — Application Layer Protocol**

Malware communicated over HTTP protocol for suspicious external communication.

- **T1059.001 — PowerShell**

Suspicious PowerShell payload (29842.ps1) was requested, indicating script-based execution behavior.

- **T1105 — Ingress Tool Transfer**

Malware attempted retrieval of external payload content from malicious infrastructure.

- **T1071.004 — DNS**

Suspicious DNS resolution activity was observed during infrastructure communication.

- **T1041 — Exfiltration Over Command and Control Channel**

Sustained outbound communication may indicate data transfer over malicious communication channels.

RISK ASSESSMENT

Based on the collected forensic evidence, the identified activity represents a high-severity cybersecurity threat. The presence of suspicious PowerShell payload retrieval, repeated outbound communication with malicious external infrastructure, suspicious DNS behavior, and sustained TCP communication strongly indicates active malware compromise.

Risk Factors Identified:

- Confirmed communication with malicious infrastructure
- Suspicious script-based payload delivery
- Possible malware persistence behavior
- Repeated outbound communication suggesting command-and-control activity
- Potential unauthorized remote access capability

Overall, Threat Level: HIGH

SOC ANALYST RECOMMENDATIONS

Based on the investigation findings, the following defensive actions are recommended to contain and mitigate the identified threat activity:

- Immediately isolate infected internal host 10.1.17.215 from the network
- Block malicious external IP address 5.252.153.241 at firewall and perimeter security controls
- Block suspicious domain infrastructure identified during DNS analysis
- Perform endpoint malware scan and forensic triage on the affected host
- Investigate PowerShell execution logs for malicious script activity
- Review proxy, firewall, and DNS logs for related indicators of compromise
- Reset credentials if credential theft or unauthorized access is suspected
- Monitor for additional communication with suspicious infrastructure across the enterprise network

CONCLUSION

This malware traffic investigation successfully identified suspicious network activity originating from internal host 10.1.17.215 through practical packet-level forensic analysis using Wireshark. Multiple indicators of malicious behavior were observed, including suspicious DNS communication, abnormal HTTP payload retrieval activity, repeated outbound communication with malicious external infrastructure, and threat intelligence confirmation through VirusTotal validation.

The evidence strongly suggests malware infection behavior involving PowerShell-based payload delivery, persistent network communication, and possible command-and-control interaction. This investigation demonstrates practical SOC analyst capabilities in malware traffic analysis, incident investigation, indicator of compromise identification, and threat intelligence correlation.

Overall, the project reflects real-world cybersecurity investigation methodology and strengthens practical blue team, digital forensics, and incident response skills relevant to professional SOC environments.